SPECIFICATION
FOR
LCM Module

MODULE No:	KD050FHFID103
CUSTOMER:	

STARTEK	INITIAL	DATE
PREPARED BY		
CHECKED BY		
APPROVED BY		

CUSTOMER	INITIAL	DATE
APPROVED BY		

Revision History

Date	Rev. No.	Page	Summary
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* Description

This is a color active matrix LTPS LCD using Low Temperature Poly-silicon TFT‘ s (Thin Film Transistors) as an active switching devices. This module is composed of a Transmissive type LTPS-LCD Panel, driver circuit, back-light unit. The resolution of a 5.0 " LTPS-LCD contains 1920X1080 pixels, and can display up to 16.7M colors.

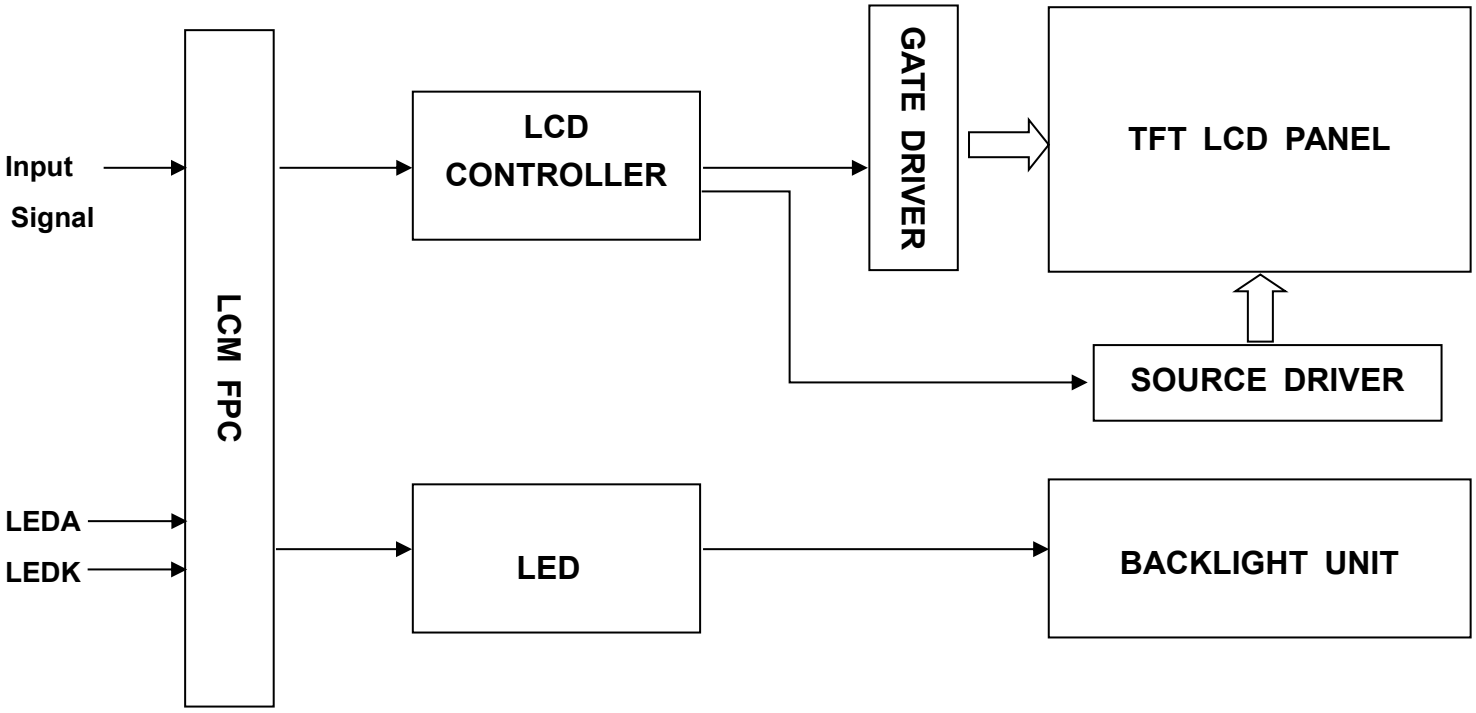
* Features

General Information Items	Specification	Unit	Note
	Main Panel		
Display area(AA)	110.592(H)*62.208(V) (5.0inch)	mm	-
Driver element	LTPS	-	-
Display colors	16.7M	colors	-
Number of pixels	1920(RGB)*1080	dots	-
Pixel arrangement	RGB vertical stripe	-	-
Pixel pitch	0.0576(H)*0.0576(V)	mm	-
Viewing angle	ALL	o'clock	-
Controller IC	SC5010E	-	-
LCM Interface	MIPI-4Lane	-	
Display mode	Transmissive/ Normally Black	-	-
Operating temperature	-20～+70	℃	-
Storage temperature	-30～+80	℃	-

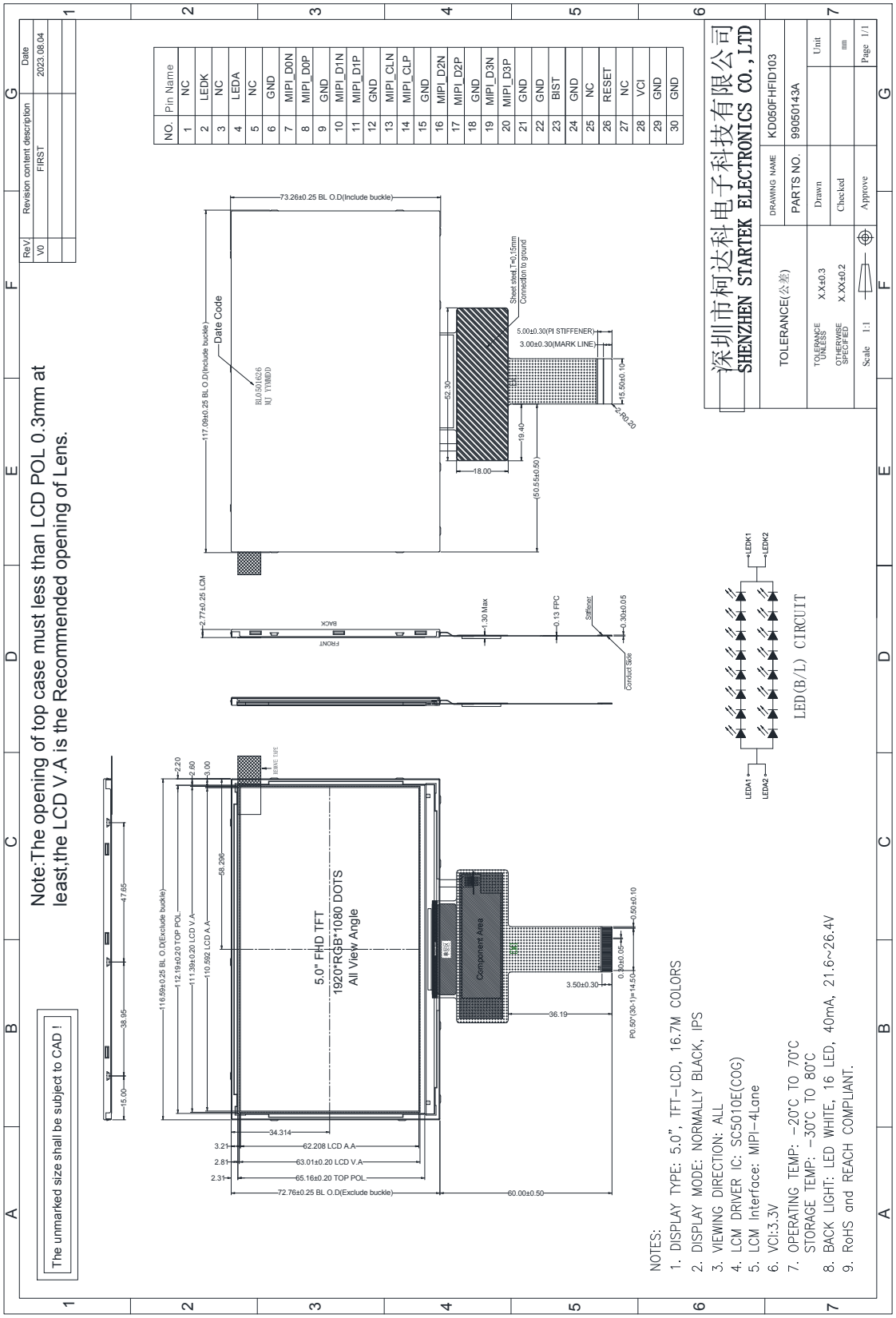
* Mechanical Information

Item		Min.	Typ.	Max.	Unit	Note
Module size	Horizontal(H)		116.59		mm	-
	Vertical(V)		72.76		mm	-
	Depth(D)		2.77		mm	-
Weight			49		g	-

1. Block Diagram



2. Outline dimension



NO.	Pin Name
1	NC
2	LEDK
3	NC
4	LEDA
5	NC
6	GND
7	MIPI_D0N
8	MIPI_D0P
9	GND
10	MIPI_D1N
11	MIPI_D1P
12	GND
13	MIPI_CLN
14	MIPI_CLP
15	GND
16	MIPI_D2N
17	MIPI_D2P
18	GND
19	MIPI_D3N
20	MIPI_D3P
21	GND
22	GND
23	BIST
24	GND
25	NC
26	RESET
27	NC
28	VCI
29	GND
30	GND

3. Input terminal Pin Assignment

NO.	SYMBOL	DISCRIPTION	I/O
1	NC	No Connection	--
2	LEDK	Cathode pin of backlight.	P
3	NC	No Connection	--
4	LEDA	Anode pin of backlight.	P
5	NC	No Connection	--
6	GND	Ground.	P
7	MIPI_D0N	- MIPI DSI differential data pair. (Data lane 0)	I
8	MIPI_D0P	Leave it open or fix to GND level when not in use.	I
9	GND	Ground.	P
10	MIPI_D1N	- MIPI DSI differential data pair. (Data lane 1)	I
11	MIPI_D1P	Leave it open or fix to GND level when not in use.	I
12	GND	Ground.	P
13	MIPI_CLN	- MIPI DSI differential clock pair	I
14	MIPI_CLP	Leave it open or fix to GND level when not in use.	I
15	GND	Ground.	P
16	MIPI_D2N	- MIPI DSI differential data pair. (Data lane 2)	I
17	MIPI_D2P	Leave it open or fix to GND level when not in use.	I
18	GND	Ground.	P
19	MIPI_D3N	- MIPI DSI differential data pair. (Data lane 3)	I
20	MIPI_D3P	Leave it open or fix to GND level when not in use.	I
21	GND	Ground.	P
22	GND	Ground.	P
23	BIST	Please open this PIN.	--
24	GND	Ground.	P
25	NC	No Connection	--
26	RESET	Reset Pin. Low active.	I
27	NC	No Connection	--
28	VCI	- Power supply (3.3V)	P
29	GND	Ground.	P
30	GND	Ground.	P

4. LCD Optical Characteristics

4.1 Optical specification

Item		Symbol	Condition	Min.	Typ.	Max.	Unit.	Note
Contrast Ratio		CR	Θ=0 Normal viewing angle	1000	1200	--		(1)(2)
Response time	Rising	$T_{R+}T_F$		--	30	40	msec	(1)(3)
	Falling							
Color gamut		S(%)		37	42	--	%	
Color Filter Chromacicity	White	W_X		-0.04	0.338	+0.04		(1) (4) CA-310
		W_Y			0.371			
	Red	R_X			0.592			
		R_Y			0.375			
	Green	G_X			0.364			
		G_Y			0.553			
	Blue	B_X			0.153			
		B_Y			0.131			
Viewing angle	Hor.	Θ_L	CR>10	75	80	--		(1)(4)
		Θ_R		75	80	--		
	Ver.	Θ_U		75	80	--		
		Θ_D		75	80	--		
Option View Direction		ALL						

*The data comes from the LCD specification.

Measuring Condition

Measuring surrounding : dark room

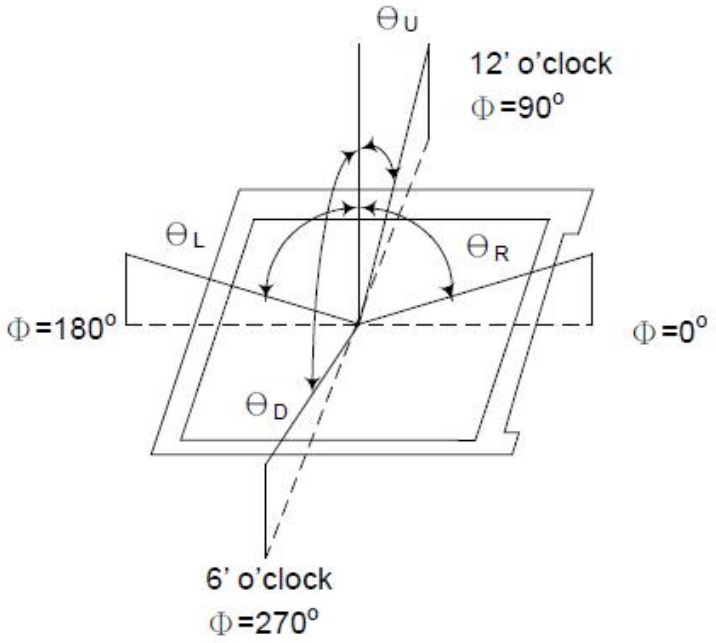
Ambient temperature : 25±2°C

15min. warm-up time.

Measuring Equipment

FPM520 of Westar Display technologies, INC., which utilized SR-3 for Chromaticity and BM-5A for other optical characteristics.

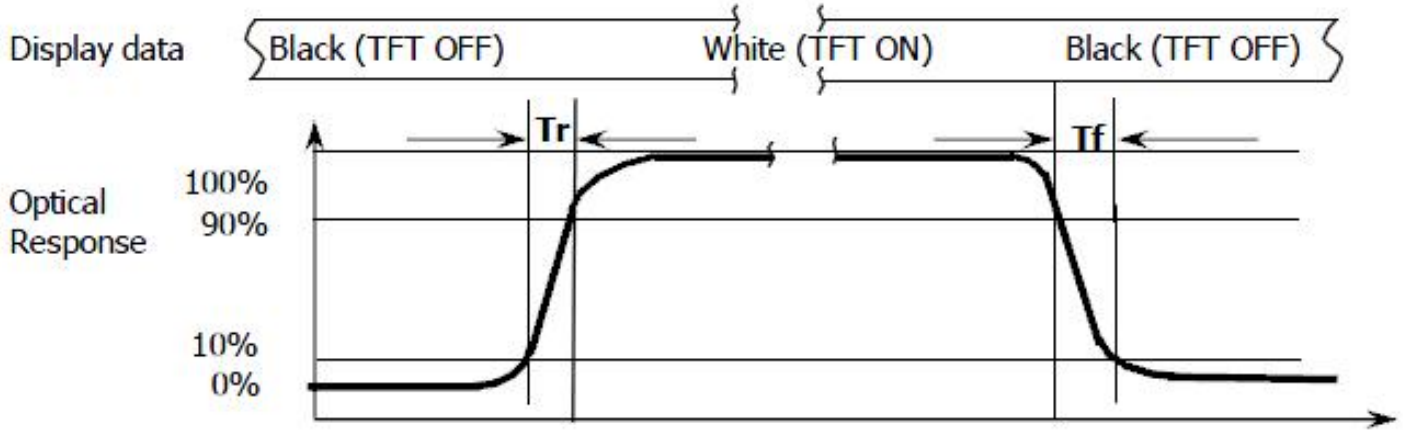
Note (1): Definition of Viewing Angle :



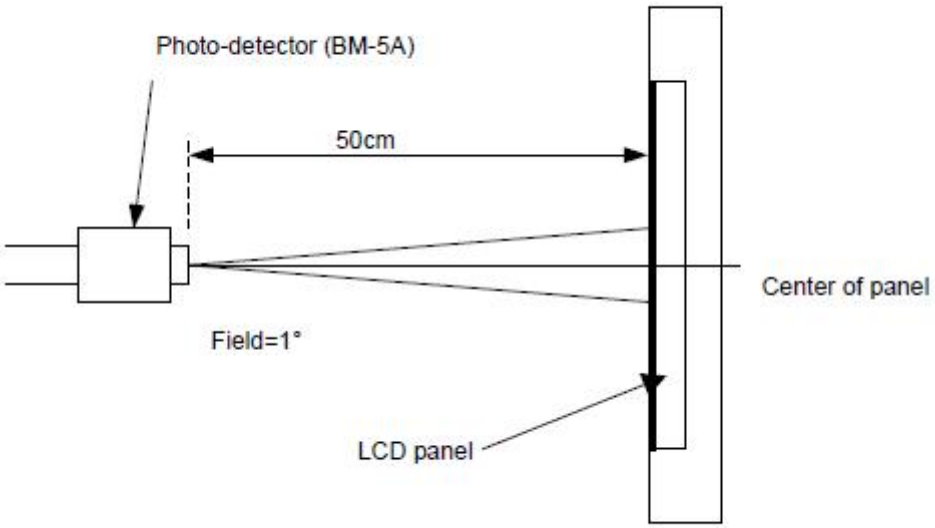
Note (2): Definition of Contrast Ratio(CR) :measured at the center point of panel

$$CR = \frac{\text{Luminance with all pixels white}}{\text{Luminance with all pixels black}}$$

Note (3): Response Time



Note (4): Definition of optical measurement setup



5. Electrical Characteristics

5.1 Absolute Maximum Rating

Characteristics	Symbol	Min.	Max.	Unit	Note
Digital Supply Voltage	VCI	-0.3	6.0	V	Note1
Operating temperature	T_{OP}	-20	+70	℃	
Storage temperature	T_{ST}	-30	+80	℃	

NOTE1: If the absolute maximum rating of even is one of the above parameters is exceeded even momentarily, the quality of the product may be degraded. Absolute maximum ratings, therefore, specify the values exceeding which the product may be physically damaged. Be sure to use the product within the range of the absolute maximum ratings.

5.2 DC Electrical Characteristics

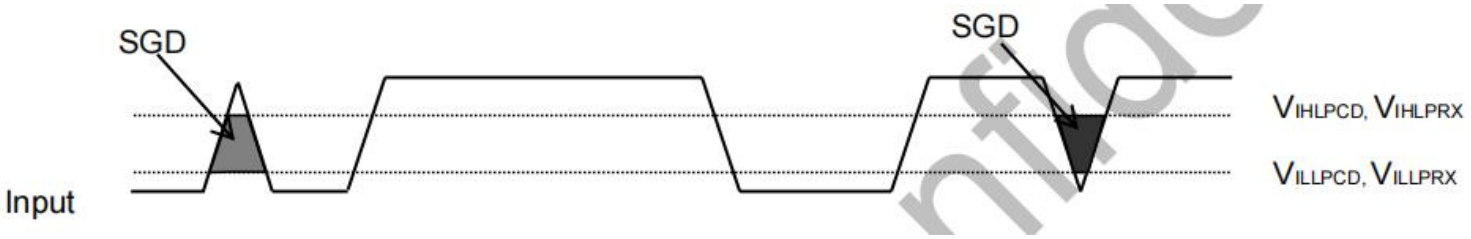
Characteristics	Symbol	Min.	Typ.	Max.	Unit	Note
Digital Supply Voltage	VCC	3.0	3.3	3.6	V	
Normal mode Current consumption	IDD	--	80	160	mA	
Level input voltage	V_{IH}	0.8VCI	--	VCI	V	
	V_{IL}	GND	--	0.2VCI	V	
Level output voltage	V_{OH}	0.8VCI	--	VCI	V	
	V_{OL}	GND	--	0.2VCI	V	

LP Mode

VSSI = VSSRX = VSSP = 0V; Temp = Ta;

Parameter	Symbol	Min	Typ.	Max.	Unit	Conditions
Logic High Level Input Voltage	V_{IHLPCD}	450	-	1350	mV	LP-CD
Logic Low Level Input Voltage	V_{ILLPCD}	0	-	200	mV	LP-CD
Logic High Level Input Voltage	V_{ILLPRX}	880	-	1350	mV	LP-RX(CLK, D0)
Logic Low Level Input Voltage	V_{ILLPRX}	0	-	550	mV	LP-RX(CLK, D0)
Logic Low Level Input Voltage	$V_{ILLPRXULP}$	0	-	300	mV	LP-RX(CLK ULP mode)
Logic High Level Input Voltage	V_{OHLPTX}	1.1	-	1.3	V	LP-TX(D0)
Logic Low Level Input Voltage	V_{OLLPTX}	-50	-	50	mV	LP-TX(D0)
Logic High Level Input Voltage	V_{IH}	-	-	10	uA	LP-CD, LP-RX
Logic Low Level Input Voltage	V_{IL}	-10	-	-	uA	LP-CD, LP-RX
Input Pulse rejection	SGD	-	-	300	Vps	DSI-CLK+/-, DSI-D0+/-

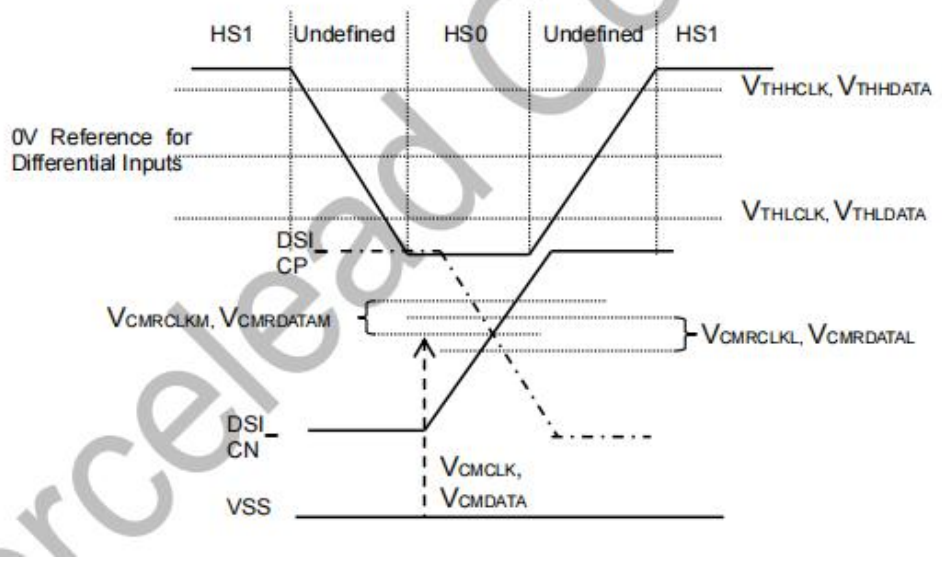
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High Speed Mode

VSSI = VSSRX = VSSP = 0V; Temp = Ta;

Parameter	Symbol	Min	Typ.	Max.	Unit	Conditions
Input Common Mode	V_{CMCLK} V_{CMDATA}	70	-	330	mV	DSI_CP/DSI_CN DSI_DxP/DSI_DxN
Input Common Mode Variation<450MHz	$V_{CMRCLKL}$ $V_{CMRCLKL}$	-50	-	50	mV	DSI_CP/DSI_CN DSI_DxP/DSI_DxN
Input Common Mode Variation>450MHz	$V_{CMRCLKM}$ $V_{CMRCLKM}$	-	-	100	mV	DSI_CP/DSI_CN DSI_DxP/DSI_DxN
Low-Level Differential Input Threshold	V_{THLCLK} $V_{THLDATA}$	-70	-	-	mV	DSI_CP/DSI_CN DSI_DxP/DSI_DxN
High-Level Differential Input Threshold	V_{THHCLK} $V_{THHDATAL}$	-	-	70	mV	DSI_CP/DSI_CN DSI_DxP/DSI_DxN
Single Ended Input Low Voltage	V_{ILHS}	-40	-	-	mV	DSI_CP/DSI_CN DSI_DxP/DSI_DxN
Single Ended Input High Voltage	V_{IHHS}	-	-	460	mV	DSI_CP/DSI_CN DSI_DxP/DSI_DxN
Differential Input Termination Resistor	R_{TERM}	80-	100	125	Ω	DSI_CP/DSI_CN DSI_DxP/DSI_DxN
Single-Ended Threshold Voltage for Termination Enable	V_{TERMEN}	-	-	450	mV	DSI_CP/DSI_CN DSI_DxP/DSI_DxN
Termination Capacitor	C_{TERM}	-	-	-	pF	DSI_CP/DSI_CN DSI_DxP/DSI_DxN



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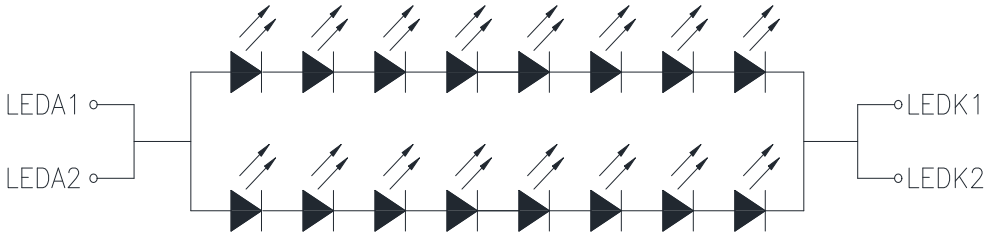
5.3 LED Backlight Characteristics

The back-light system is edge-lighting type with 16 chips LED

Item	Symbol	Min.	Typ.	Max.	Unit	Note
Forward Current	I_F	30	40	--	mA	
Forward Voltage	V_F	21.6	24	26.4	V	
LCM Luminance (IF=40mA)	LV	850	1000	--	cd/m2	Note3
LED life time	Hr	50000	--	--	Hour	Note1,2
Uniformity	Avg	80	--	--	%	Note3

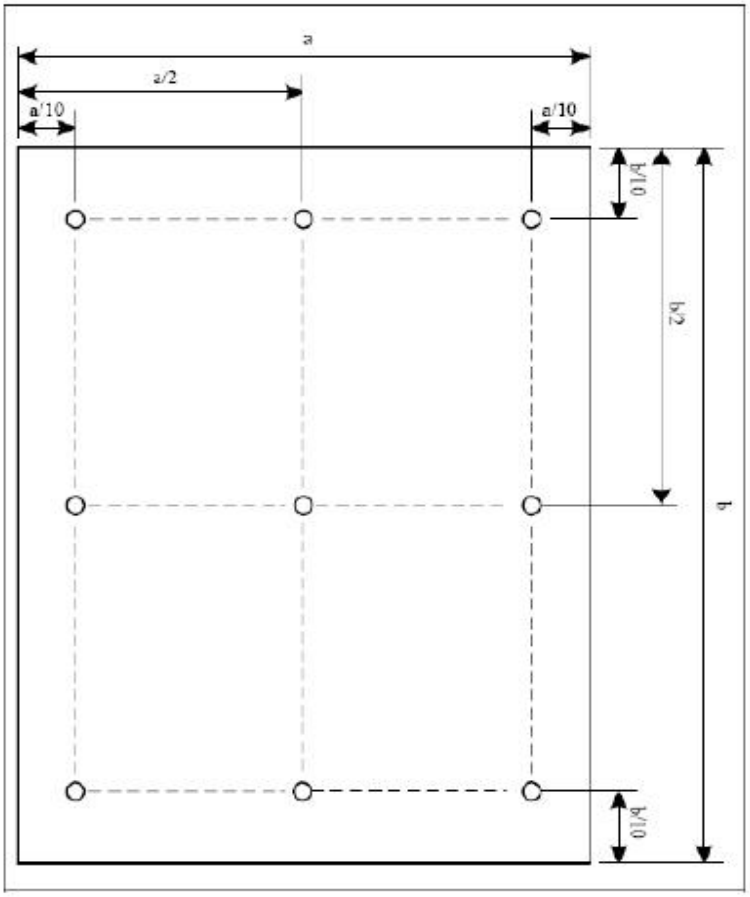
Note1: LED life time (Hr) can be defined as the time in which it continues to operate under the condition: Ta=25±3 ℃, typical IL value indicated in the above table until the brightness becomes less than 50%.

Note 2: The “LED life time” is defined as the module brightness decrease to 50% original brightness at Ta=25℃ and IL=40mA. The LED lifetime could be decreased if operating IL is larger than 40mA. The constant current driving method is suggested.



LED(B/L) CIRCUIT

Note (3) Luminance Uniformity of these 9 points is defined as below:



$$\text{Uniformity} = \frac{\text{minimum luminance in 9 points (1-9)}}{\text{maximum luminance in 9 points (1-9)}}$$

$$\text{Luminance} = \frac{\text{Total Luminance of 9 points}}{9}$$

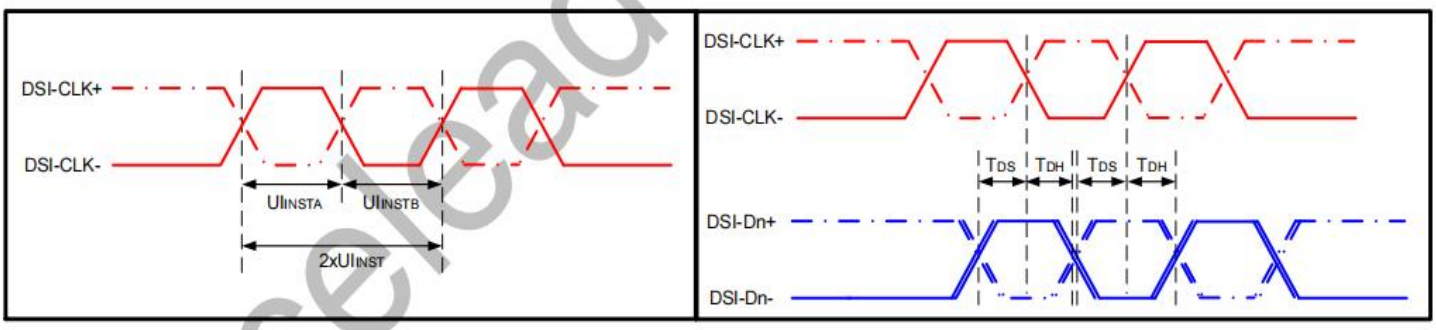
6. AC Characteristic

6.1 MIPI AC Characteristics

VSSI = VSSRX = VSSP = 0V, VDDI = VDDP= VDDRX = 3.0 ~ 3.3V, Ta Temperature

Parameter	Symbol	Min	Typ.	Max.	Unit	Conditions
Double UI Instantaneous	$2xUI_{INSTA}$	2.4	-	25	ns	DSI-CLK+/-
UI Instantaneous Half	UI_{INSTA} UI_{INSTB}	1.2	-	12.5	ns	DSI-CLK+/-
Data to Clock Setup Time	T_{DS}	0.15	-	-	UI	DSI-Dn+/-
Data to Clock Hold Time	T_{DH}	0.15	-	-	UI	DSI-Dn+/-

Note. This chip supports MIPI maximum frequency 840Mbps data rate.



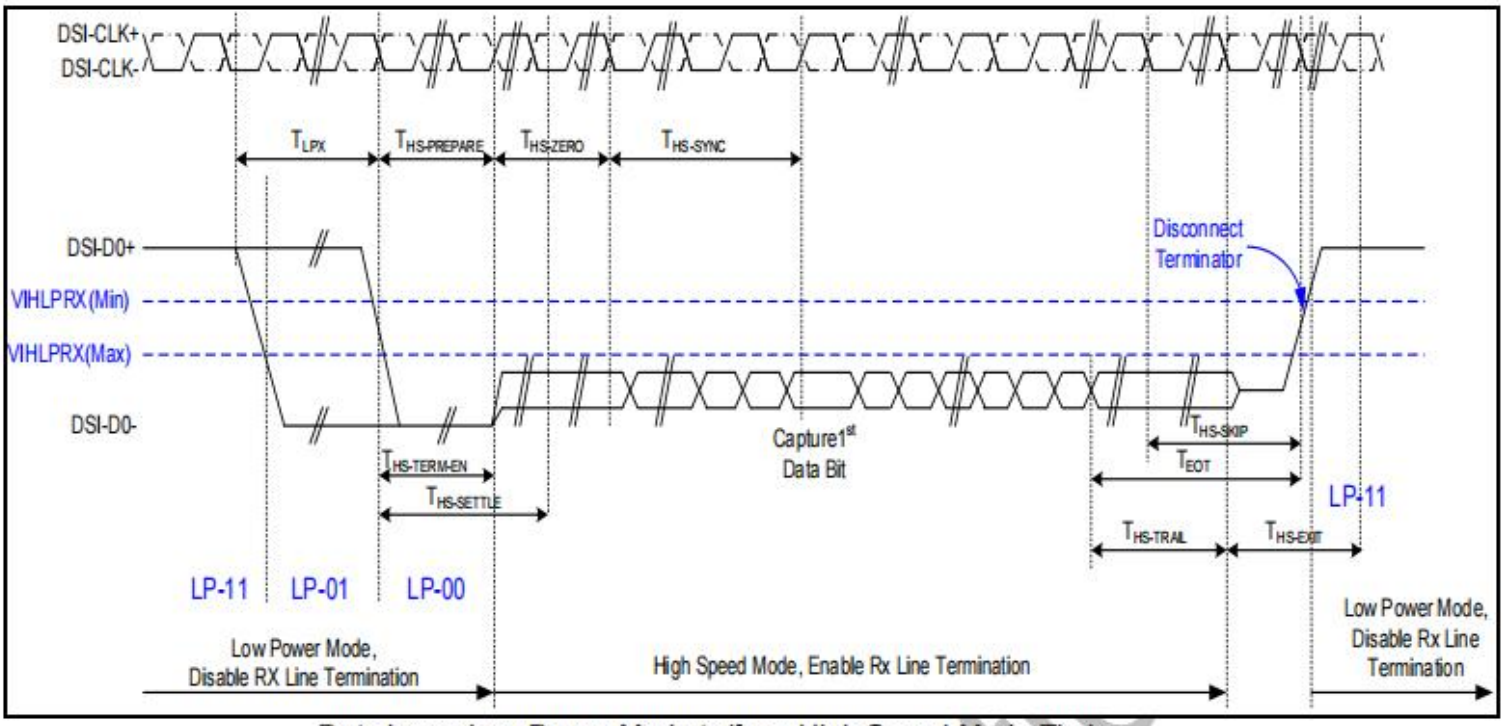
Clock Channel Timing

VSSI = VSSRX = VSSP = 0V, VDDI = VDDP= VDDRX = 3.0 ~ 3.3V, Ta Temperature

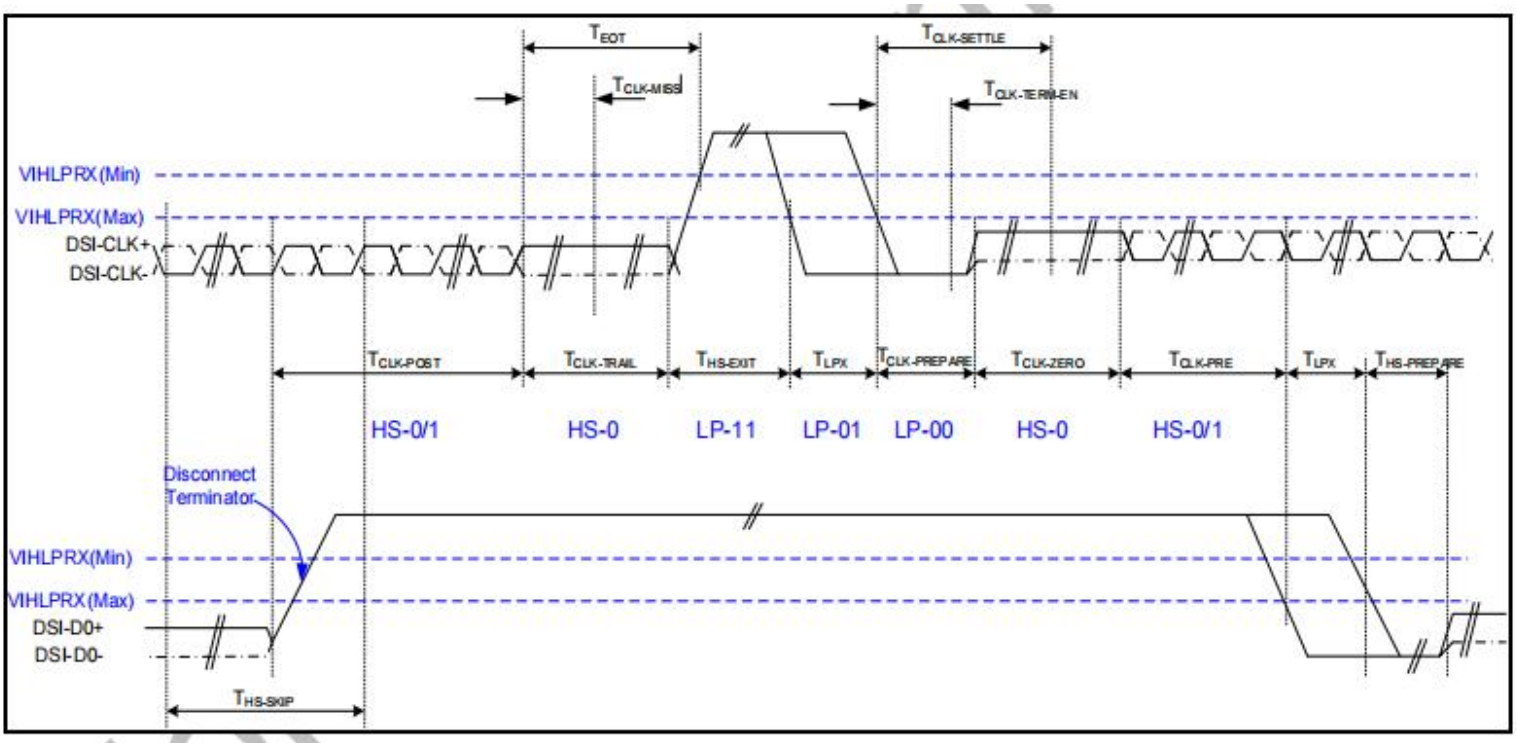
Signal	Symbol	Parameter	MIN	MAX	Unit	Description
Low Power Mode to High Speed Mode Timing						
DSI-Dn+/-	TLPX	Length of any low power state period	50	-	ns	Input
DSI-Dn+/-	THS-PREPARE	Time to drive LP-00 to prepare for HS transmission	40+4UI	85+6UI	ns	Input
DSI-Dn+/-	THS-TERM-EN	Time to enable data receiver line termination measured from when Dn crosses VILMAX	-	35+4UI	ns	Input
DSI-Dn+/-	THS-PREPARE + THS-ZERO	THS-PREPARE + time to drive HS-0 before the sync sequence	140+10UI	-	ns	Input
High Speed Mode to Low Power Mode Timing						
DSI-Dn+/-	THS-SKIP	Time-out at display module to ignore transition period of EoT	40	55+4UI	ns	Input
DSI-Dn+/-	THS-EXIT	Time to drive LP-11 after HS burst	100	-	ns	Input
DSI-Dn+/-	THS-TRAIL	Time to drive flipped differential state after last payload data bit of a HS transmission burst	60+4UI	-	ns	Input
High Speed Mode to/from Low Power Mode Timing						
DSI-CLK+/-	TCLK-POS	Time that the MPU shall continue sending HS clock after the last associated data lane has transition to LP mode	60+52UI	-	ns	Input

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DSI-CLK+/-	TCLK-TRAIL	Time to drive HS differential state after last payload clock bit of a HS transmission burst	60	-	ns	Input
DSI-CLK+/-	THS-EXIT	Time to drive LP-11 after HS burst	100	-	ns	Input
DSI-CLK+/-	TCLK-PREPARE	Time to drive LP-00 to prepare for HS transmission	38	95	ns	Input
DSI-CLK+/-	TCLK-TERM-EN	Time-out at clock lan display module to enable HS transmission	--	38	ns	Input
DSI-CLK+/-	TCLK-PREPARE + TCLK-ZERO	Minimum lead HS-0 drive period before starting clock	300	-	ns	Input
DSI-CLK+/-	TCLK-PRE	Time that the HS clock shall be driven prior to any associated data lane beginning the transition from LP to HS mode	8UI	-	ns	Input
DSI-CLK+/-	TEOT	Time form start of TCLK-TRAIL period to start of LP-11 state	-	105ns+12UI	ns	Input



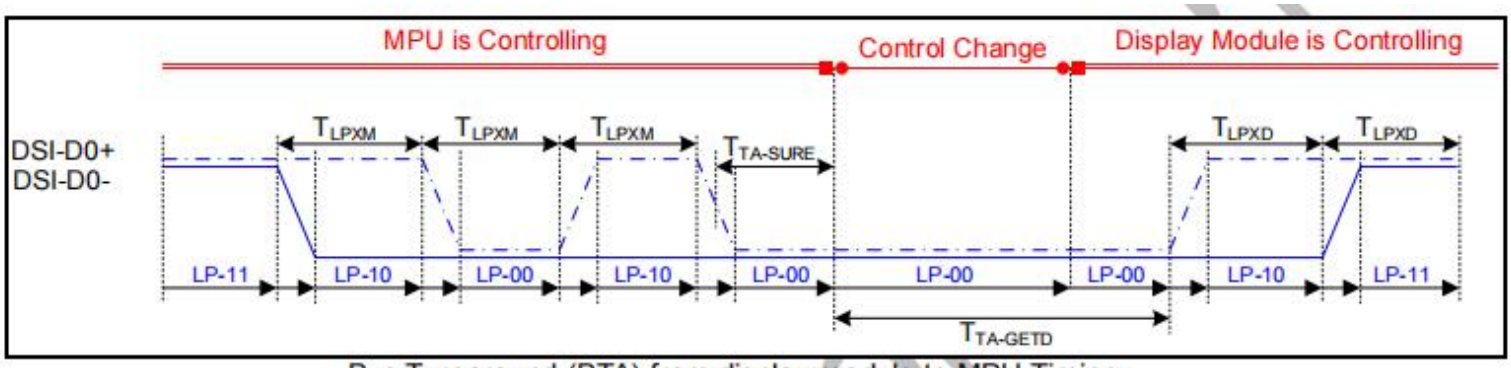
Data lanes-Low Power Mode to/from High Speed Mode Timing



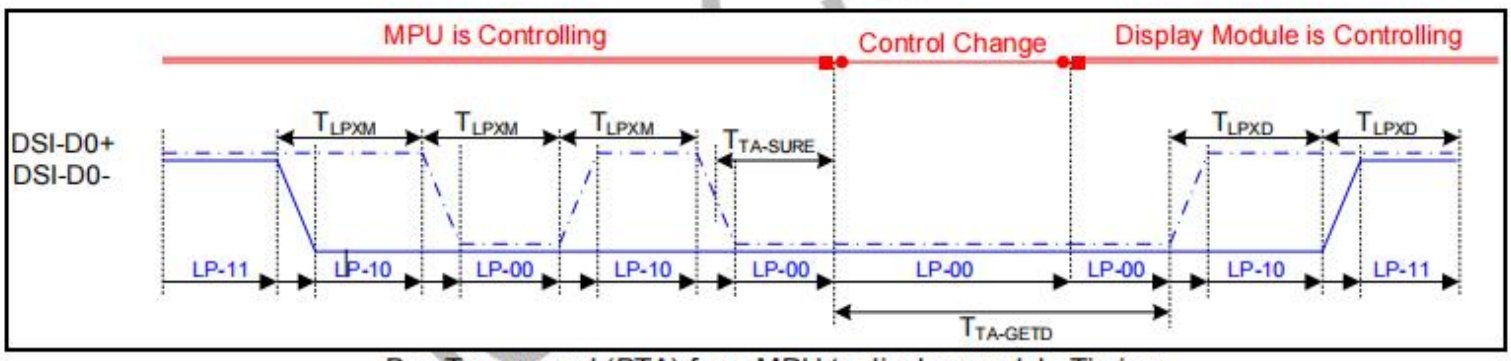
Clock lanes- High Speed Mode to/from Low Power Mode Timing

VSSI = VSSRX = VSSP = 0V, VDDI = VDDP= VDDRX = 3.0 ~ 3.3V, Ta Temperature

Signal	Symbol	Parameter	MIN	MAX	Unit	Description
DSI-D0+/-	TLPXM	Length of LP-00,LP-01, LP-10 or LP-11 periods MPU→Display Module	50	75	ns	Input
DSI-D0+/-	TLPXD	Length of LP-00,LP-01, LP-10 or LP-11 periods MPU→Display Module	50	75	ns	Output
DSI-D0+/-	TTA-SURED	Time-out before the MPU start driving	T_{LPXD}	$2xT_{LPXD}$	ns	Output
DSI-D0+/-	TTA-GETD	Time to drive LP-00 by display module	$5xT_{LPXD}$		ns	Input
DSI-D0+/-	TTA-GOD	Time to drive LP-00 after turnaround request-MPU	$4xT_{LPXD}$		ns	Output

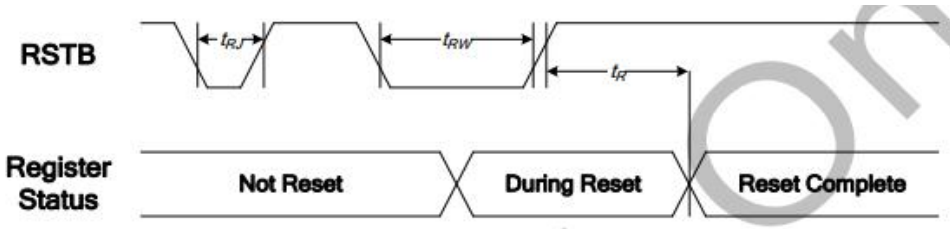


Bus Turnaround (BTA) from display module to MPU Timing



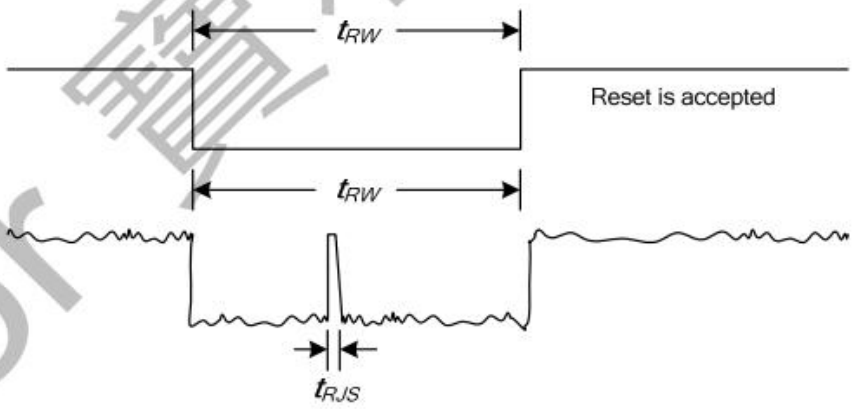
Bus Turnaround (BTA) from MPU to display module Timing

6.2 Hardware Reset Timing



VSSI = VSSRX = VSSP = 0V, VDDI = VDDP= VDDRX = 3.0 ~ 3.3V, Ta = -40 ~ 85°C

Item	Signal	Symbol	Condition	Rating		Unit
				Min.	Max.	
Reset time	RSTB	t_R		—	5	us
Reset "L" pulse width		t_{RW}		15	—	
Reset rejection		t_{RJ}		—	5	
Reset rejection (for noise spike)		t_{RJS}		—	10	ns



Note:

1. For PROM related operation, it takes 50ms at least for PROM Registers to load PROM contents. Do not use any PROM related command during this period.
2. When the system issues a RSTB low pulse, the reset procedure of IC will start if the low pulse is longer than t_{RW} specified above. If the low pulse is less than t_{RJ} specified above, the reset procedure of IC will not start. If the low pulse is longer than t_{RJ} and less than t_{RW}, the reset procedure of IC is not guaranteed.

For 量产 Only

7. LCD Module Out-Going Quality Level

7.1 VISUAL & FUNCTION INSPECTION STANDARD

7.1.1 Inspection conditions

Inspection performed under the following conditions is recommended.

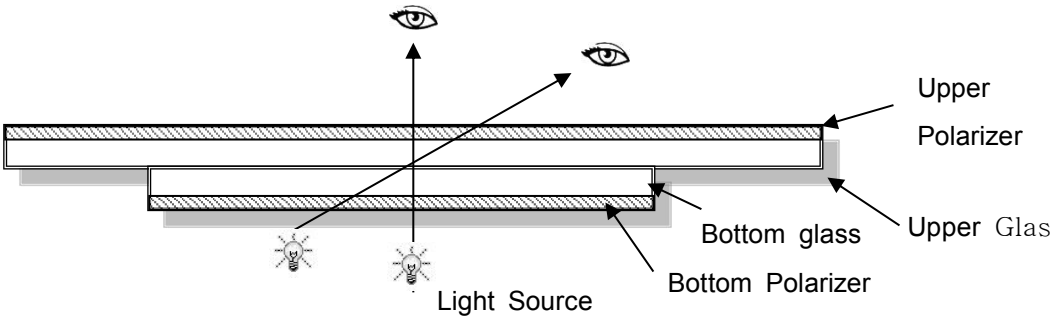
Temperature : 25±5℃

Humidity : 65%±10%RH

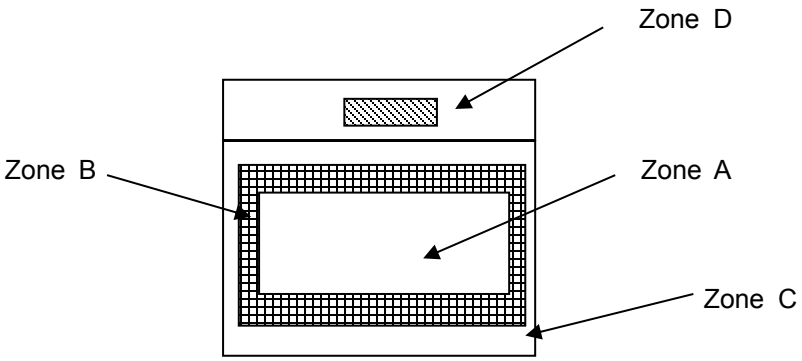
Viewing Angle : Normal viewing Angle.

Illumination: Single fluorescent lamp (300 to 700Lux)

Viewing distance:30-50cm



7.1.2 Definition



Zone A : Effective Viewing Area(Character or Digit can be seen)

Zone B : Viewing Area except Zone A

Zone C : Outside (Zone A+Zone B) which can not be seen after assembly by customer .)

Zone D : IC Bonding Area

Note:As a general rule ,visual defects in Zone C can be ignored when it doesn't effect product function or appearance after assembly by customer

7.1.3 Sampling Plan

According to GB/T 2828-2003 ; , normal inspection, Class Ⅱ

AQL:

Major defect	Minor defect
0.65	1.5

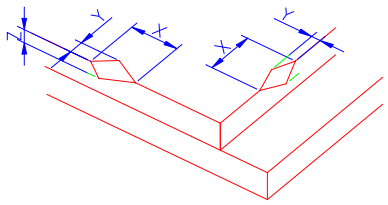
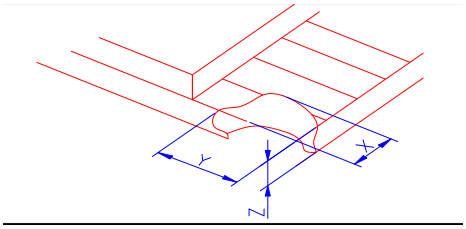
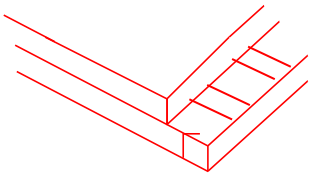
LCD: Liquid Crystal Display , LCM: Liquid Crystal Module, CTP: Capacitive Touch Panel

No	Items to be inspected	Criteria	Classification of defects
1	Functional defects	1) No display, Open or miss line 2) Display abnormally, Short 3) Backlight no lighting, abnormal lighting. etc	Major
2	Missing	Missing components and etc	
3	Outline dimension	Overall outline dimension beyond the drawing is not allowed,deformation and etc	
4	Color tone	Color unevenness, refer to limited sample	Minor
5	Spot/Line defect	Light dot,Dim spot,(Note1) Polarizer Air Bubble, Polarizer accidented spot and etc	
6	Soldering appearance	Good soldering , Peeling off is not allowed and etc	
7	LCD/Polarizer/CTP	Black/White spot/line, scratch, crack, etc.	

Note1: a) Light dot：Dots appear bright and unchanged in size in which LCD panel is displaying under black pattern.

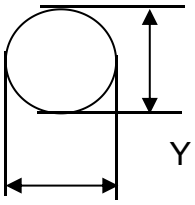
b) Dim dot: Dots appear dark and unchanged in size in which LCD panel is displaying under pure red, green, blue picture.

7.1.4 Criteria (Visual)

<table>
<tr><th>Number</th><th>Items</th><th>Criteria(mm)</th></tr>
<tr><td rowspan="3">1.0 LCD Crack/Broken
NOTE:
X: Length
Y: Width
Z: Height
L: Length of ITO,
T: Height of LCD</td><td>(1) The edge of LCD broken</td><td> 
<table><tr><th>X</th><th>Y</th><th>Z</th></tr><tr><td>≤3.0mm</td><td><Inner border line of the seal</td><td>≤T</td></tr></table></td></tr>
<tr><td>(2)LCD corner broken</td><td> 
<table><tr><th>X</th><th>Y</th><th>Z</th></tr><tr><td>≤3.0mm</td><td>≤L</td><td>≤T</td></tr></table></td></tr>
<tr><td>(3) LCD crack</td><td> 
Crack
Not allowed</td></tr>
</table>

2.0

Spot defect



Φ=(X+Y)/2

① light dot（ black/white spot , pinhole, stain，etc.）

Zone / Size (mm)	Acceptable Qty		
	A	B	C
Φ≤0.15	Ignore	Ignore	
0.15<Φ≤0.25	3(distance≧10mm)		
0.25<Φ≤0.4	2(distance≧10mm)		
Φ>0.4	0		

② Dim spot（light leakage、dent、dark spot，etc）

Zone / Size (mm)	Acceptable Qty		
	A	B	C
Φ≤0.15	Ignore	Ignore	
0.15<Φ≤0.25	3(distance≧10mm)		
0.25<Φ≤0.4	2(distance≧10mm)		
Φ>0.4	0		

③ Polarizer accidented spot

Zone / Size (mm)	Acceptable Qty		
	A	B	C
Φ≤0.2	Ignore		Ignore
0.2<Φ≤0.5	2(distance≧10mm)		
Φ>0.5	0		

④Polarizer Bubble

Zone / Size (mm)	Acceptable Qty		
	A	B	C
Φ≤0.2	Ignore		Ignore
0.2<Φ≤0.4	2(distance≧10mm)		
Φ>0.4	0		

3.0 LCD Pixel defect

Pixel bad points

Item	Zone A	Acceptable Qt
Bright dot	Random	N≤2
	2 dots adjacent	N≤0
	3 dots adjacent	N≤0
Dark dot	Random	N≤3
	2 dots adjacent	N≤0
	3 dots adjacent	N≤0
Distance	1. Minimum Distance Between Bright dots. 2. Minimum Distance Between dark dots 3. Minimum Distance Between dark and bright dot.	5mm
Total bright and dark dot		N≤4

Note:

A) Bright dot：Dots appear bright and unchanged in size in which LCD panel is displaying under black pattern.

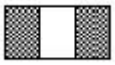
B) Dark dot: Dots appear dark and unchanged in size in which LCD panel is displaying under pure red, green, blue picture.

C) 2 dot adjacent = 1 pair = 2 dots

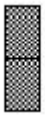
Picture:



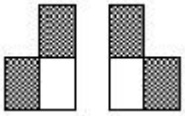
2 dot adjacent



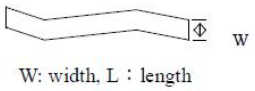
2 dot adjacent



2 dot adjacent (vertical)



2 dot adjacent (slant)

<table>
<tr>
<td>4.0</td>
<td>Line defect (LCD /Polarizer backlight black/white line, scratch, stain)
 

N：Count</td>
<td>
<table>
<tr><td rowspan="2">Width(mm)</td><td rowspan="2">Length(mm)</td><td colspan="3">Acceptable Qty</td></tr>
<tr><td>A</td><td>B</td><td>C</td></tr>
<tr><td>Φ≤0.05</td><td>Ignore</td><td colspan="2">Ignore</td><td rowspan="3">Ignore</td></tr>
<tr><td>0.05<W≤0.06</td><td>L≤5.0</td><td colspan="2">N≤3</td></tr>
<tr><td>0.06<W≤0.08</td><td>L≤4.0</td><td colspan="2">N≤2</td></tr>
<tr><td>W>0.08</td><td colspan="4">Define as spot defect</td></tr>
</table>
</td>
</tr>
<tr>
<td>5.0</td>
<td>Electronic Components SMT.</td>
<td>Not allow missing parts，solderless connection，cold solder joint，mismatch，The positive and negative polarity opposite</td>
</tr>
<tr>
<td>6.0</td>
<td>Display color& Brightness.</td>
<td>1. Color：Measuring the color coordinates, The measurement standard according to the datasheet or samples.
2. Brightness：Measuring the brightness of White screen, The measurement standard according to the datasheet or Samples.</td>
</tr>
<tr>
<td>7.0</td>
<td>LCD Mura/Waving/ Hot spot</td>
<td>Not visible through 5% ND filter in 50% gray or judge by limit sample if necessary.</td>
</tr>
</table>

Criteria (functional items)

Number	Items	Criteria (mm)
1	No display	Not allowed
2	Missing segment	Not allowed
3	Short	Not allowed
4	Backlight no lighting	Not allowed
5	CTP no function	Not allowed

8. Reliability Test Result

Item	Condition	Inspection after test
High Temperature Operating	70℃,96H	Inspection after 2~4hours storage at room temperature, the sample shall be free from defects: 1.Air bubble in the LCD; 2.Non-display; 3.Missing segments/line; 4.Glass crack; 5.Current IDD is twice higher than initial value.
Low Temperature Operating	-20℃, 96HR	
High Temperature Storage	80℃, 96HR	
Low Temperature Storage	-30℃, 96HR	
High Temperature & High Operating	+60℃, 90% RH ,96 hours.	
Thermal Shock (Non-operation)	-10℃,30 min ↔60℃,30 min, Change time:5min 20CYC.	
ESD test	C=150pF, R=330,5points/panel Air:±8KV, 5times; Contact:±6KV, 5 times; (Environment: 15℃~35℃, 30%~60%).	
Vibration (Non-operation)	Frequency range:10~55Hz, Stroke:1.5mm Sweep:10Hz~55Hz~10Hz 2 hours for each direction of X.Y.Z. (6 hours for total) (Package condition).	
Box Drop Test	1 Corner 3 Edges 6 faces,80㎝(MEDIUM BOX)	

Remark:

1.The test samples should be applied to only one test item.

2.Sample size for each test item is 5~10pcs.

3.For Damp Proof Test, Pure water(Resistance＞10MΩ) should be used.

4.In case of malfunction defect caused by ESD damage, if it would be recovered to normal state after resetting, it would be judged as a good part.

5.Failure Judgment Criterion: Basic Specification, Electrical Characteristic, Mechanical Characteristic, Optical Characteristic.

6. The color fading mura of polarizing filter should not care.

9. Cautions and Handling Precautions

9.1 Handling and Operating the Module

(1) When the module is assembled, it should be attached to the system firmly.
Do not warp or twist the module during assembly work.
(2) Protect the module from physical shock or any force. In addition to damage, this may cause improper operation or damage to the module and back-light unit.
(3) Note that polarizer is very fragile and could be easily damaged. Do not press or scratch the surface.
(4) Do not allow drops of water or chemicals to remain on the display surface.
If you have the droplets for a long time, staining and discoloration may occur.
(5) If the surface of the polarizer is dirty, clean it using some absorbent cotton or soft cloth.
(6) The desirable cleaners are water, IPA (Isopropyl Alcohol) or Hexane.
Do not use ketene type materials (ex. Acetone), Ethyl alcohol, Toluene, Ethyl acid or Methyl chloride. It might permanent damage to the polarizer due to chemical reaction.
(7) If the liquid crystal material leaks from the panel, it should be kept away from the eyes or mouth. In case of contact with hands, legs, or clothes, it must be washed away thoroughly with soap.
(8) Protect the module from static; it may cause damage to the CMOS ICs.
(9) Use finger-stalls with soft gloves in order to keep display clean during the incoming inspection and assembly process.
(10) Do not disassemble the module.
(11) Protection film for polarizer on the module shall be slowly peeled off just before use so that the electrostatic charge can be minimized.
(12) Pins of I/F connector shall not be touched directly with bare hands.
(13) Do not connect, disconnect the module in the “Power ON” condition.
(14) Power supply should always be turned on/off by the item 6.1 Power On Sequence &6.2 Power Off Sequence

9.2 Storage and Transportation.

(1) Do not leave the panel in high temperature, and high humidity for a long time.
It is highly recommended to store the module with temperature from 0 to 35 ℃ and relative humidity of less than 70%
(2) Do not store the TFT-LCD module in direct sunlight.
(3) The module shall be stored in a dark place. When storing the modules for a long time, be sure to adopt effective measures for protecting the modules from strong ultraviolet radiation, sunlight, or fluorescent light.
(4) It is recommended that the modules should be stored under a condition where no condensation is allowed. Formation of dewdrops may cause an abnormal operation or a failure of the module.
In particular, the greatest possible care should be taken to prevent any module from being operated where condensation has occurred inside.

(5) This panel has its circuitry FPC on the bottom side and should be handled carefully in order not to be stressed.

10. Packing

----TBD----------